Birth Defects

By

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TW Sadler. Birth defects: In Langman's medical embryology, 15th edition,2024, Chapter 9: P130-141.

Learning objectives

- Terminology.
- Causes.
- Genetic factors.
- Principles of teratology.
- Environmental factors.

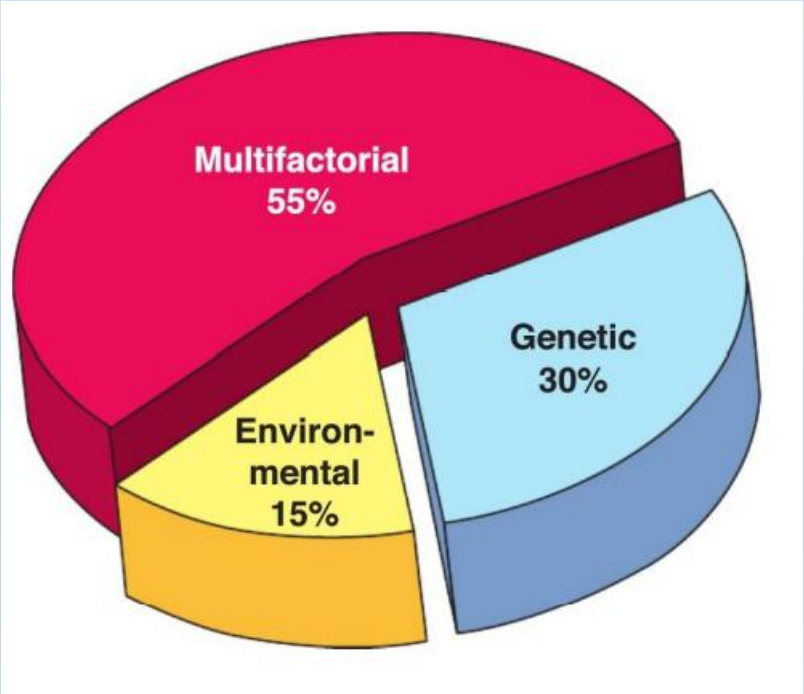
Birth defect, **congenital malformation**, and **congenital anomaly** are synonymous terms used to describe **structural**, **behavioral**, **functional**, and **metabolic** disorders present **at birth**.

The study of these disorders is **Teratology** (Gr. teratos; monster) or **Dysmorphology**.

Teratologist **or** **Dysmorphologist** are usually within the department of **Clinical genetics**.

The causes of birth defects fall into three categories:

- Genetic factors (30%).
- Environmental factors (15%).
- Interaction of the environment with a person's genetic susceptibility (55%).



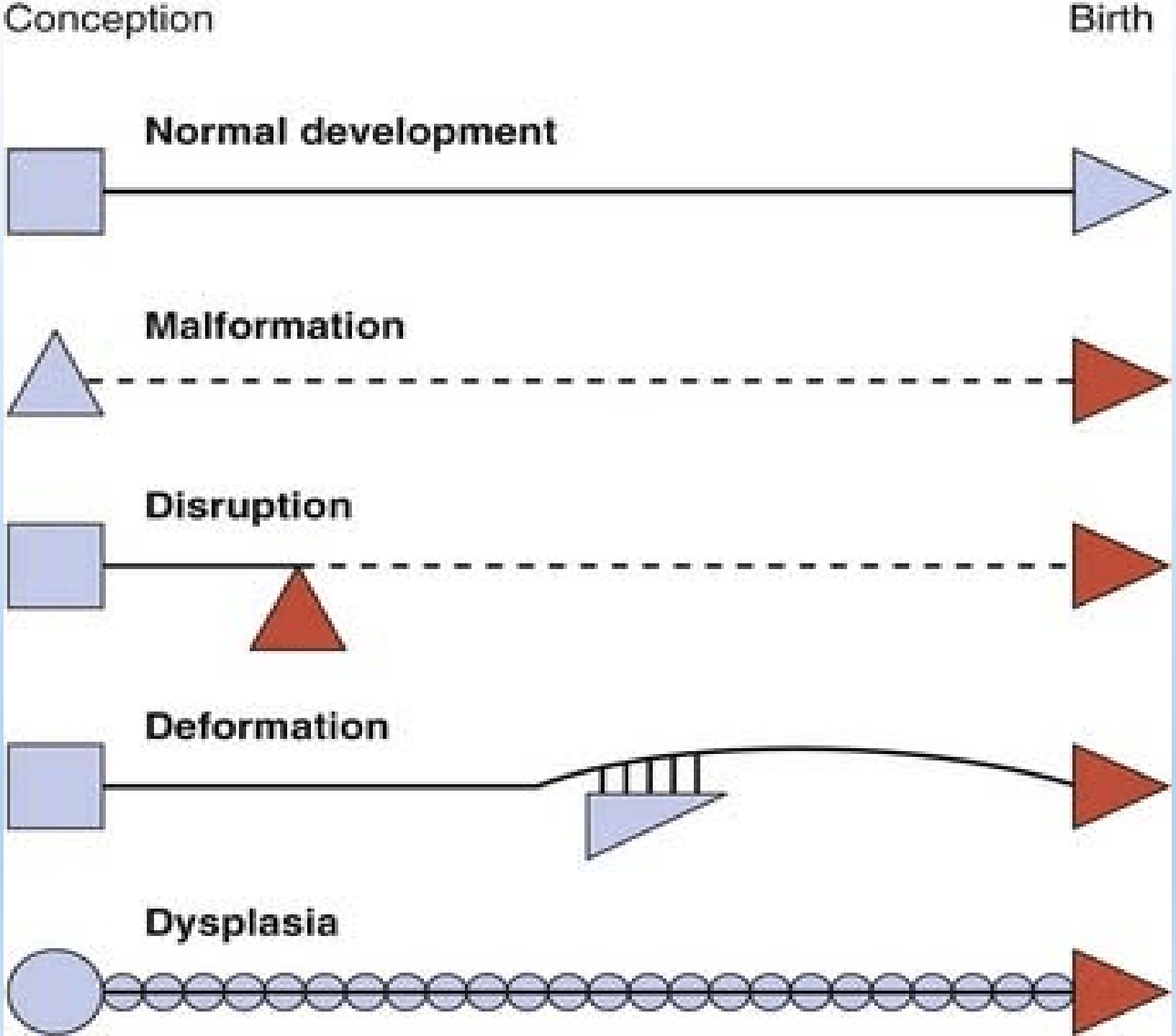
Major structural anomalies occur in approximately 3% of live born infants.

A leading cause of infant mortality, accounting for approximately 25% of infant deaths.

They are **nondiscriminatory**; the frequencies of birth defects are the same for Asians, African Americans, Latin Americans, Whites, and Native Americans.

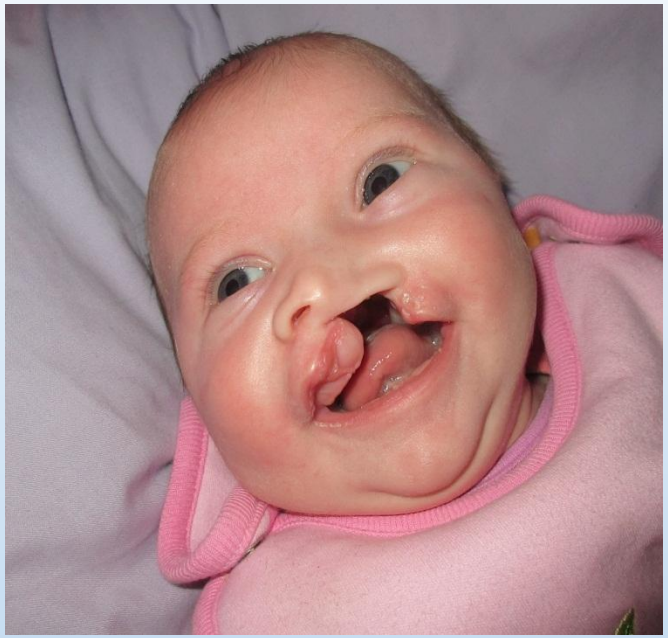
- Minor anomalies occur in approximately 15% o f newborns.
- Infants with one minor anomaly have a 3% chance of having a major malformation.
- Infants with two minor anomalies have a 10% chance of having a major malformation.
- Infants with three or more minor anomalies have a 20% chance of having a major malformation.
- Minor anomalies like microtia (small ears), are indicators for the presence of major malformations.



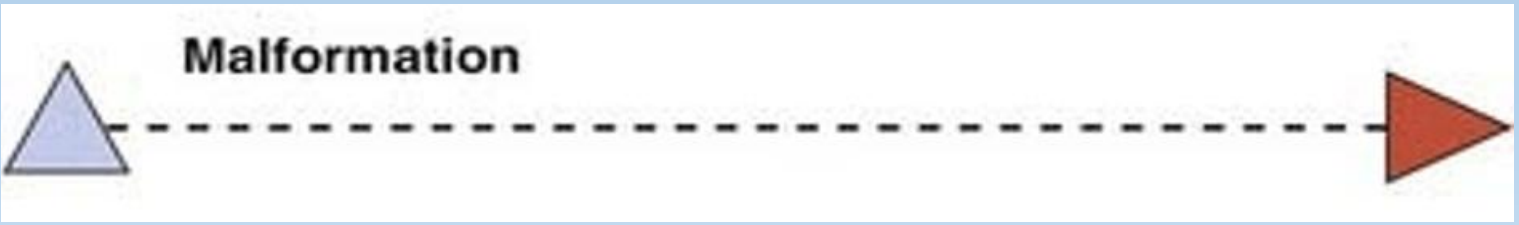

Conception
Birth
Normal development
Malformation
Disruption
Deformation
Dysplasia

Malformations

- Occur during formation of structures and organs (embryonic period).
- They are primary or intrinsic.
- They may result in complete or partial absence of a structure or organ.
- Malformations are caused by environmental and/or genetic factors acting independently or in concert.
- VSD, ASD, cleft lip/palate, neural tube defect.





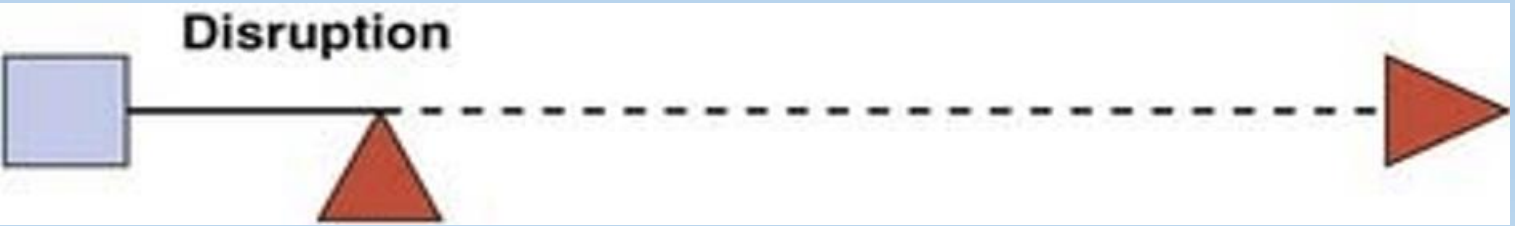


Disruptions

- Result in morphological alterations of already formed structures.
- Caused by destructive processes.
- They are secondary or extrinsic.
- Transverse limb defects and finger amputation produced by amniotic bands.
- Mostly caused by environmental factors.

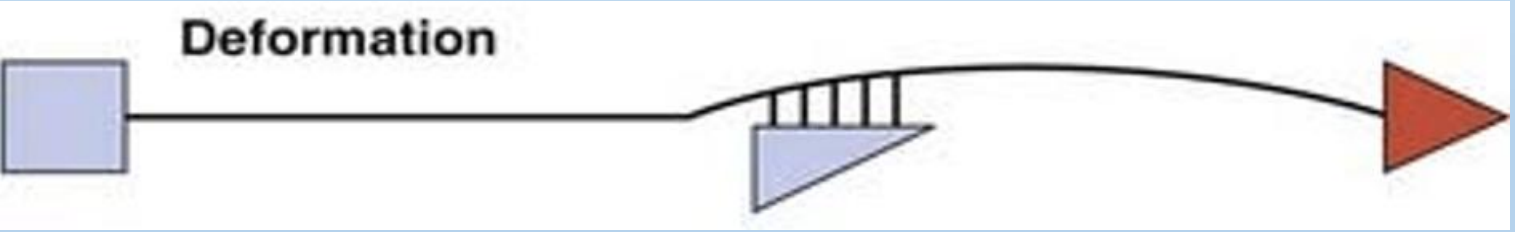


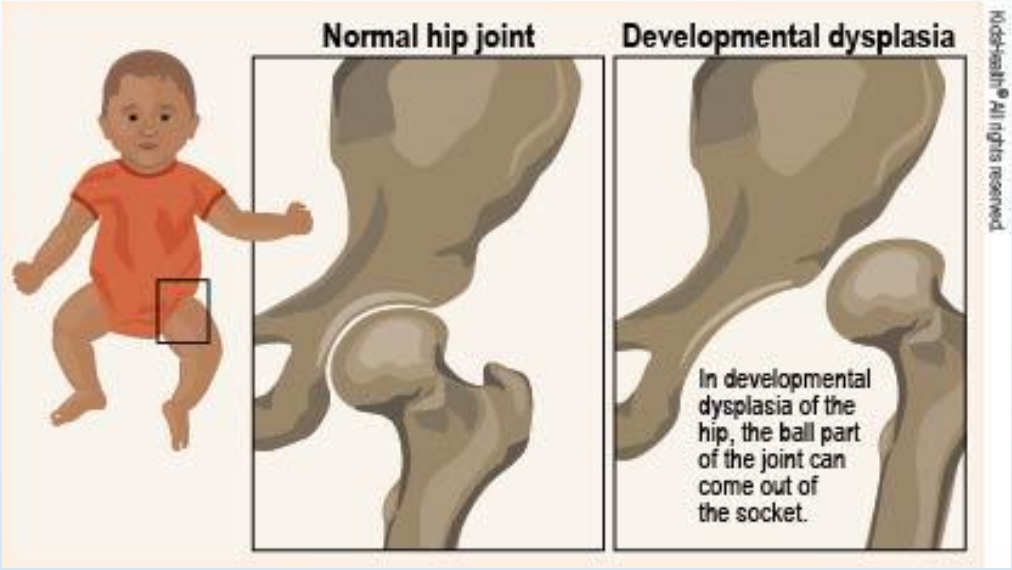




Deformations

- Result from mechanical forces that deform an otherwise normal part of the fetus.
- Deformations often involve the musculoskeletal system and may be reversible postnatally.
- CDH, TEV dt oligohydramnios are caused by compression in the amniotic cavity.



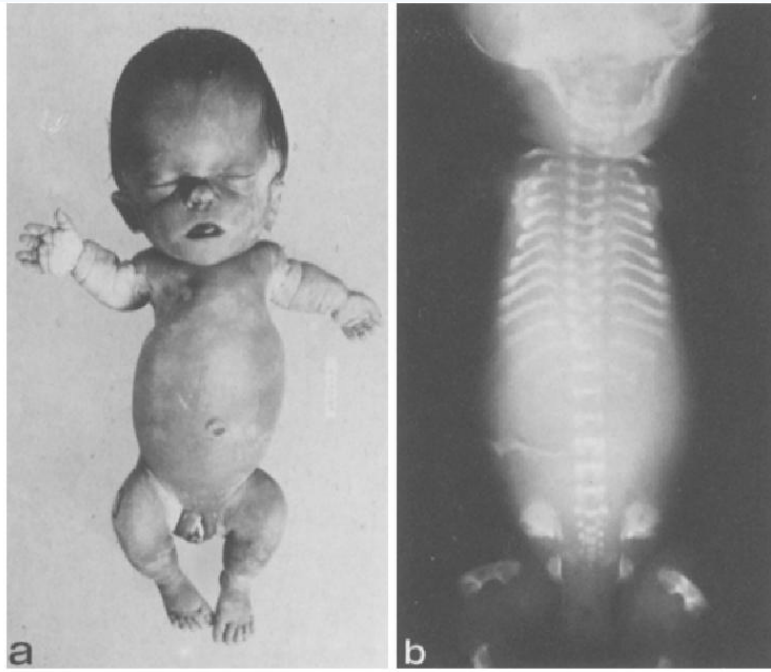




Dysplasia

- Abnormal organization or assembly of specific cells into tissue.
- The effects are seen wherever that particular tissue is present.
- Skeletal dysplasia caused by mutant FGFR3 gene.
- Ectodermal dysplasia.
- Mostly caused by a single gene defect.
- High recurrence rate.







Syndrome

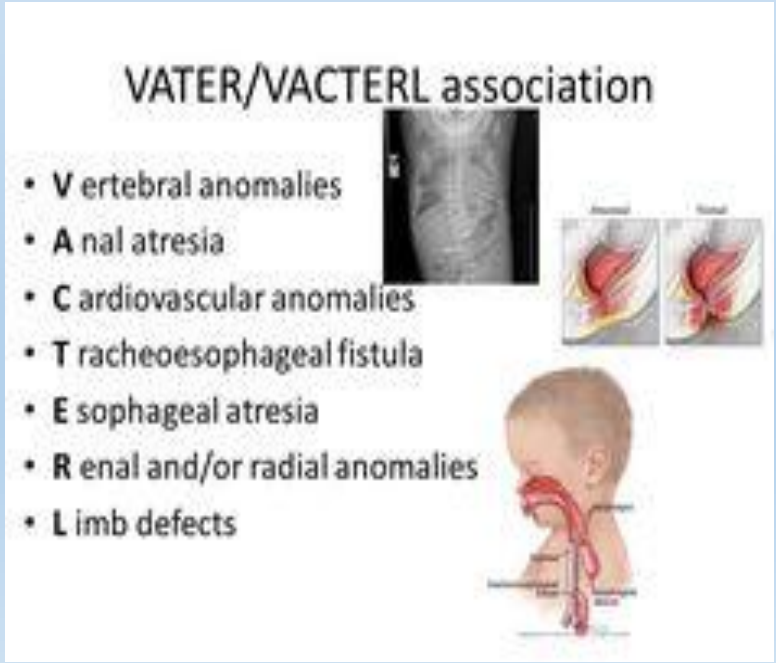
- Is a group of anomalies occurring together that have a specific common cause.
- The cause is known and the risk of recurrence is known.
- Caused by chromosomal defect as Down syndrome.
- Or single gene (IRF-6) defect as Van der Woude syndrome.



Figure 1: Bilateral and symmetrical lower lip pit with unilateral complete cleft lip and palate.

Association

- Is the non-random appearance of two or more anomalies that occur together more frequently than by chance alone.
- But the cause has not been determined.
- An example is the VACTERL association (vertebral, anal, cardiac, tracheoesophageal, renal, and limb anomalies).
- Associations are important because recognition of one or more of the components promotes the search for other anomalies.



Sequence

- A single primary factor cause a cascade of events.
- Potter sequence, oligohydramnios due to chronic leakage of amniotic fluid or defective fetal urine output.
- Leads to fetal compression, flattened face, CDH, TEV, lung hypoplasia and death from respiratory failure.



Genetic causes of malformations

- 1-Chromosomal abnormalities.
- 2-Single-Gene defect.
- 3-Multifactorial inheritance.
- 4-Genetic heterogeneity.

Chromosomal abnormalities

- Common chromosomal syndromes.



Down syndrome

Trisomy 18 - Edwards Syndrome

- 2nd most common trisomy
- 80% female
- Majority die before birth
- Low survival rate
  - heart abnormalities, kidney malformations, etc.
- Symptoms: small head , malformed ears, widely spaced eyes, clenched hands

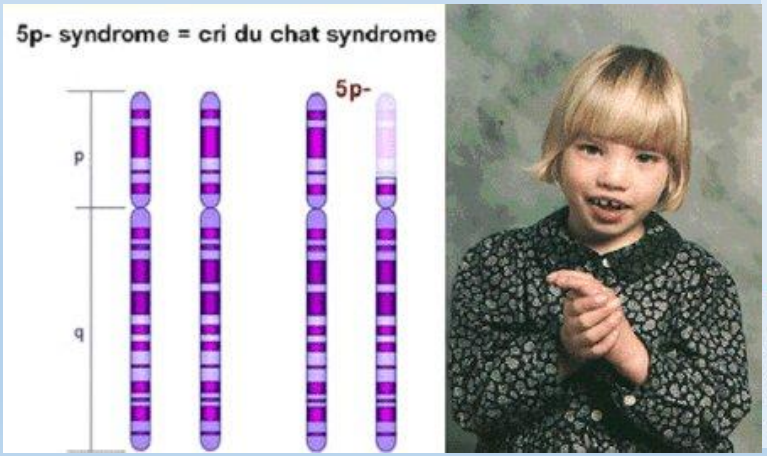


Chromosomal abnormalities

- **Microdeletion syndromes.**



Di George syndrome 22q11



Cri-du-chat syndrome 5p

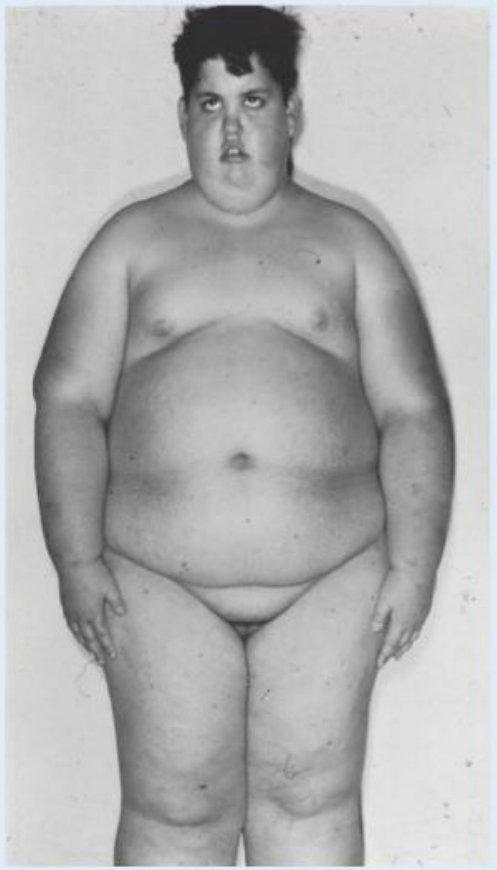


FIGURE 2.14 Patient with Prader-Willi syndrome resulting from a microdeletion on paternal chromosome 15. If the defect is inherited on the maternal chromosome, Angelman syndrome occurs (Fig. 2.13).

Prader-Willi syndrome



FIGURE 2.13 Patient with Angelman syndrome resulting from a microdeletion on maternal chromosome 15. If the defect is inherited on the paternal chromosome, Prader-Willi syndrome occurs (Fig. 2.14).

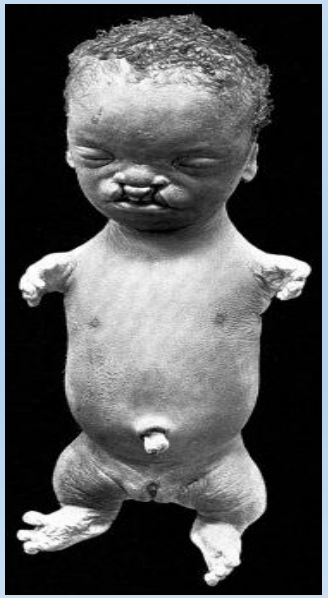
Angelman syndrome

Single-Gene defect

- Isolated: affect one organ or one system: hydrocephalus, microcephaly, aniridia, cataract and polydactyly.
- Syndromes: Apert (FGFR2), Ectrodactyly Ectodermal Dysplasia, Roberts, Van der Woude syndrome.







Gene over-expression

- **Dicephalus.**
- Caused by over- expression of **GOOSE-COID gene**.



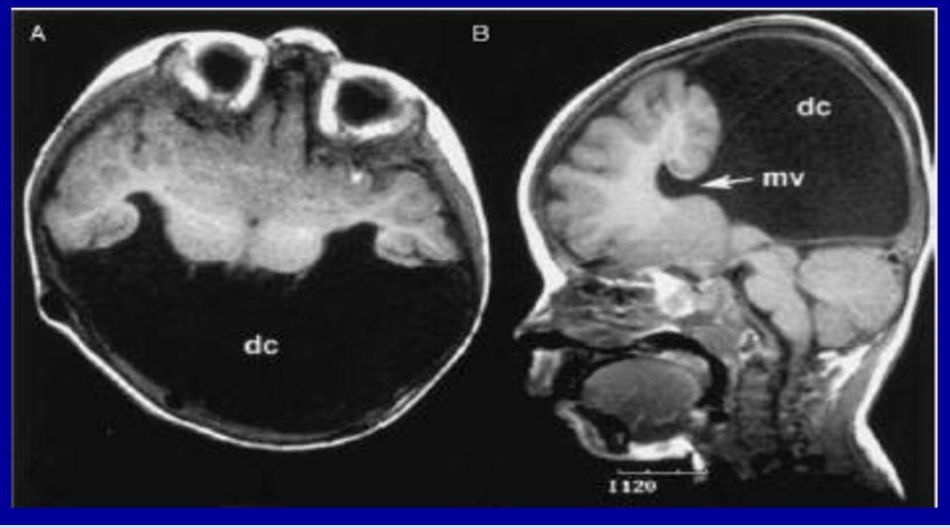
Multifactorial inheritance

- Accounts for the majority of congenital malformation due to genetic causes.
- As in Type 1 DM: inheritance involves many loci (polygenic).

Genetic heterogeneity

- Specific congenital malformations have many different causes.
- Chromosomal, syndrome or isolated.
- Holoprosencephaly, cleft lip/palate and NTDs.

Alobar HPE. (A) lack of separation of the two hemispheres. Large dorsal cyst (dc) posteriorly. (B) reveals a midline ventricle, a monoventricle (mv), that communicates posteriorly with the dorsal cyst (dc).



Environmental Factors

- Until the early 1940s, it was assumed that congenital malformations were caused primarily by hereditary factors.
- N. Gregg discovered that **rubella** affecting a mother during early pregnancy can cause abnormalities in the embryo.
- The sedative **thalidomide** made it clear that drugs could also cross the placenta and produce birth defects.
- Since that time, many agents have been identified as **teratogens**.

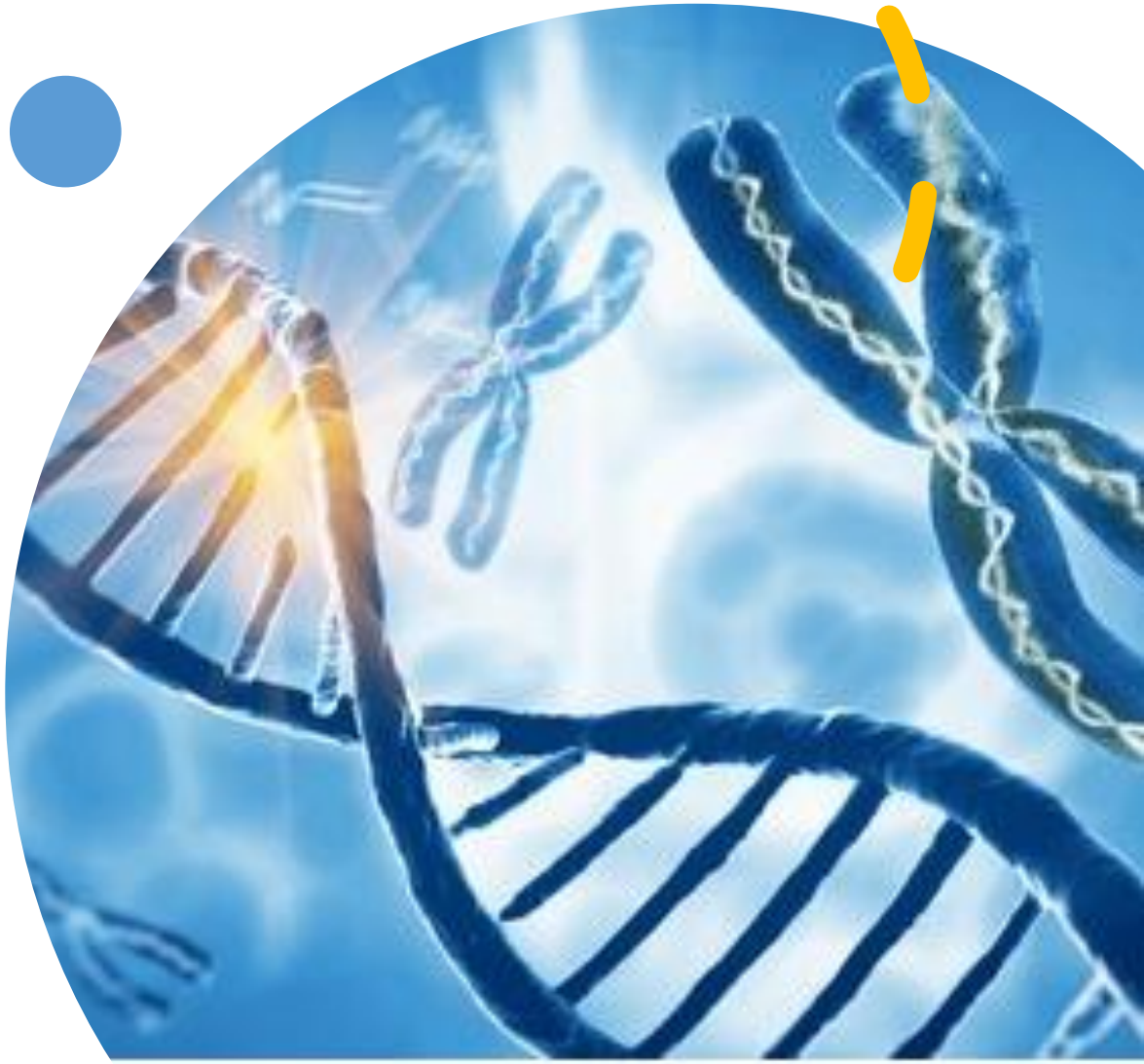


Principles of Teratology

Factors determining the capacity of an agent to produce birth defects:

1. Genotype of the fetus and how this genetic composition interacts with the environment.

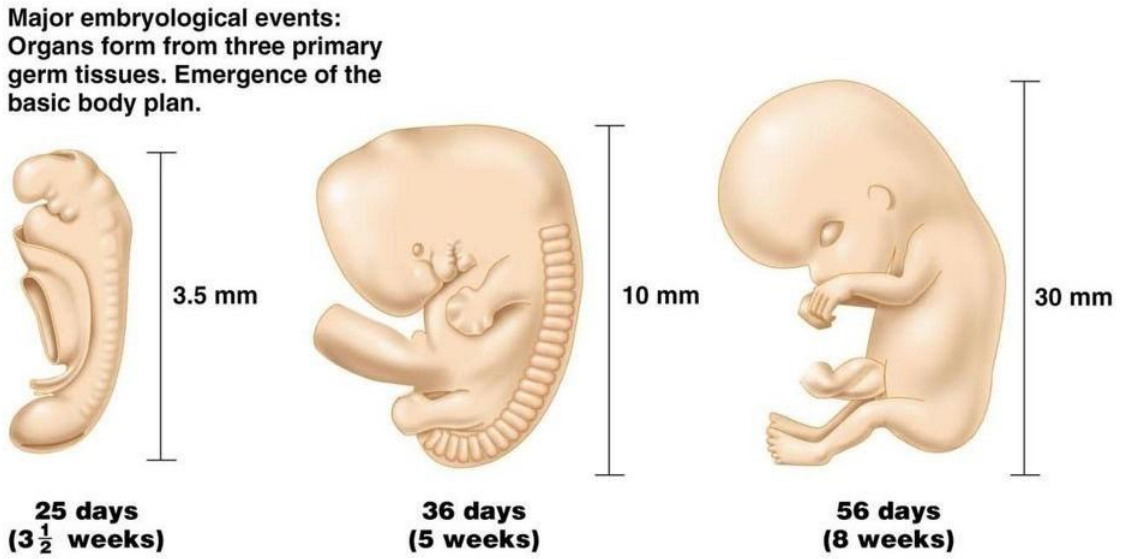
- The maternal genome is also important and affects biochemical and molecular processes that affect the fetus.



Principles of Teratology

2. Developmental stage at the time of exposure and susceptibility to teratogens.

- The most sensitive period for inducing birth defects is the third to eighth weeks of gestation (embryonic period).
- **No stage of development is completely safe.**



Principles of Teratology

3. Manifestations of abnormal development depend on dose and duration of exposure to a teratogen.

Principles of Teratology

4. Teratogens act in specific ways (mechanisms) on developing cells and tissues to initiate abnormal embryogenesis (pathogenesis).

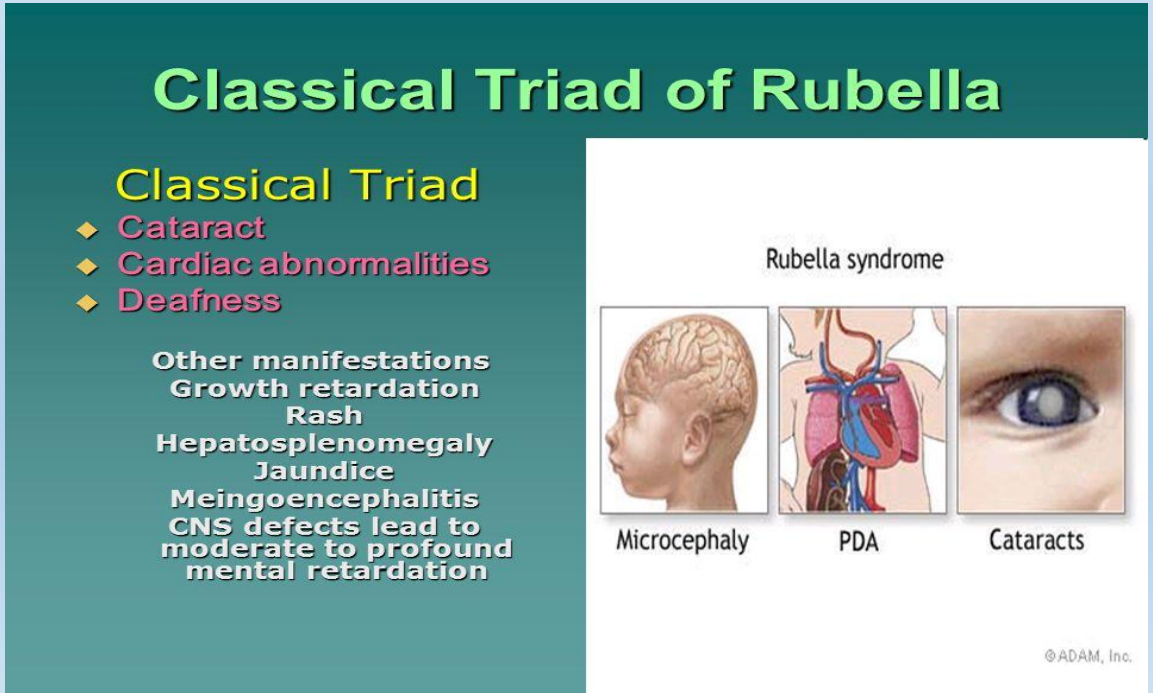
- Mechanisms may involve inhibition of a specific biochemical or molecular process, cell death, decreased cell proliferation, or other cellular phenomena.

Principles of Teratology

5. Manifestations of abnormal development could be death, malformation, intrauterine growth restriction, and functional disorders.

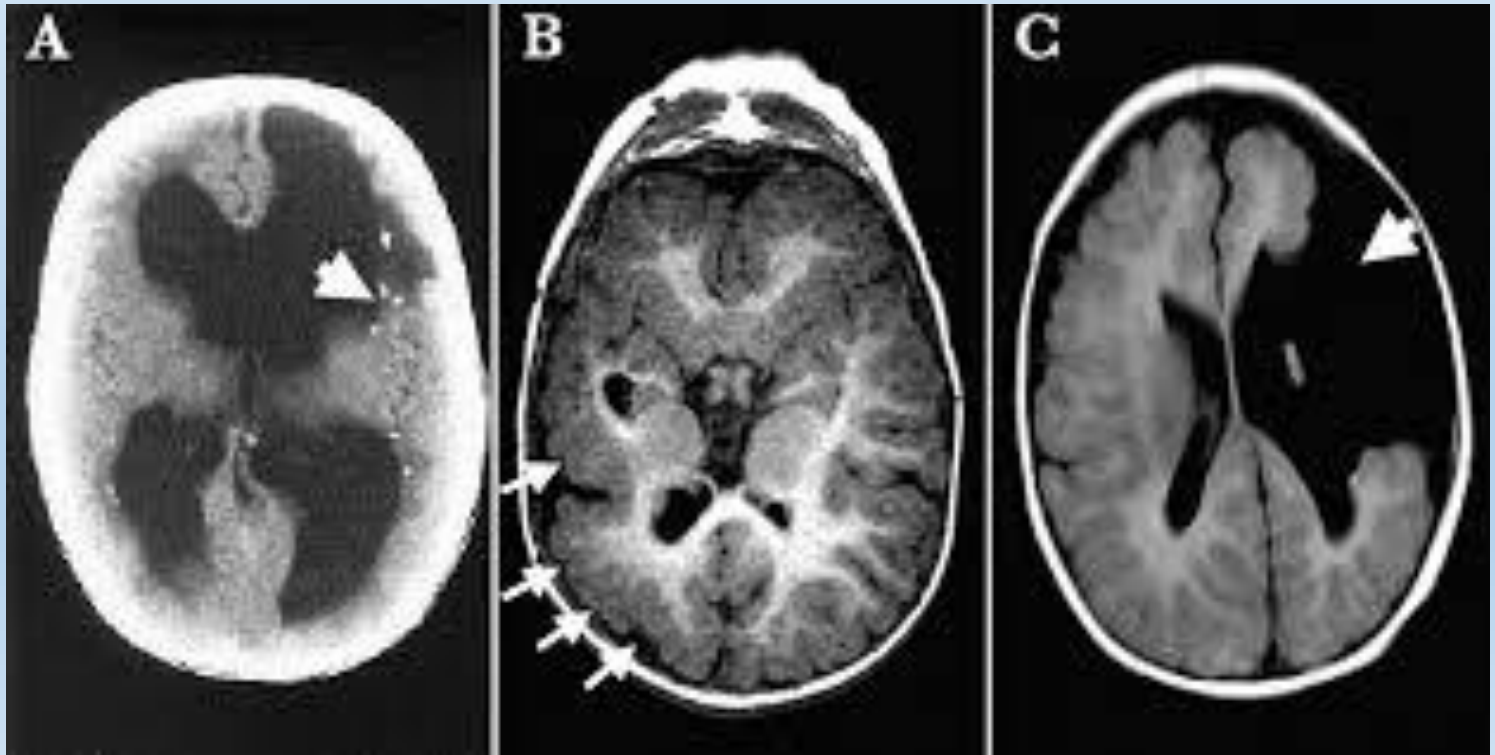
Infectious Agents

- Infectious agents that cause birth defects include a number of viruses.
- Birth defects due to rubella (German measles) during pregnancy (**congenital rubella syndrome**) used to be a major problem.
- Cataracts, glaucoma, heart defects (PDA), hearing loss, microcephaly and tooth abnormalities.
- Development and widespread use of vaccine have nearly eliminated congenital malformations from this cause.





- **Cytomegalovirus** is a serious threat.
- Often, the mother has no symptoms, but the effects on the fetus can be devastating.
- The infection can cause serious illness at birth as microcephaly and is sometimes fatal.
- On the other hand, some infants are asymptomatic at birth but develop abnormalities later, including hearing loss, visual impairment, and intellectual disability.



- **Herpes simplex virus** can rarely cause birth defects (in embryonic period): Microphthalmia, microcephaly, retinal dysplasia.
- Usually, infection is transmitted to the child during delivery, causing severe illness and sometimes death.



- **Herpes zoster virus (varicella)** causes scarring of the skin, limb hypoplasia, defects of the eyes and central nervous system.
- Infants whose mothers are infected before 13 weeks gestation: 0.4% are malformed,
- Infants whose mothers are infected during 13 to 20 weeks of gestation: 2% are malformed.
- Infection at the time of labor can lead to severe neonatal infection and fatal meningoencephalitis.



Figure 2: Close-up view of the scar over the left forearm

- **Zika virus:**
- Congenital Zika syndrome: Microcephaly



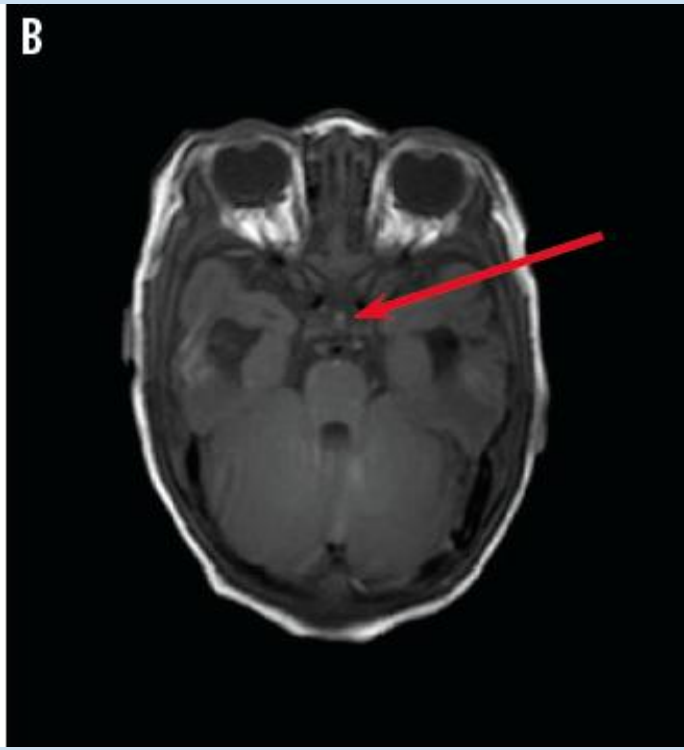
- **Toxoplasmosis** can cause birth defects.
- Poorly cooked meat; feces of domestic animals, especially cats can carry the parasite Toxoplasmosis gondii.
- A characteristic feature of fetal toxoplasmosis infection is cerebral calcifications and hydrocephalus.
- Infants may appear normal at birth but later can develop visual impairment, hearing loss, seizures, and intellectual disability.

Congenital toxoplasmosis



Clinical manifestations. Hydrocephalus, lesions in the organs of the vision (chorioretinitis), cirrosis of the liver and enlargement of the spleen.





Physical agents

X-rays:

Microcephaly, spina bifida, cleft palate, limb defects.

Hyperthermia:

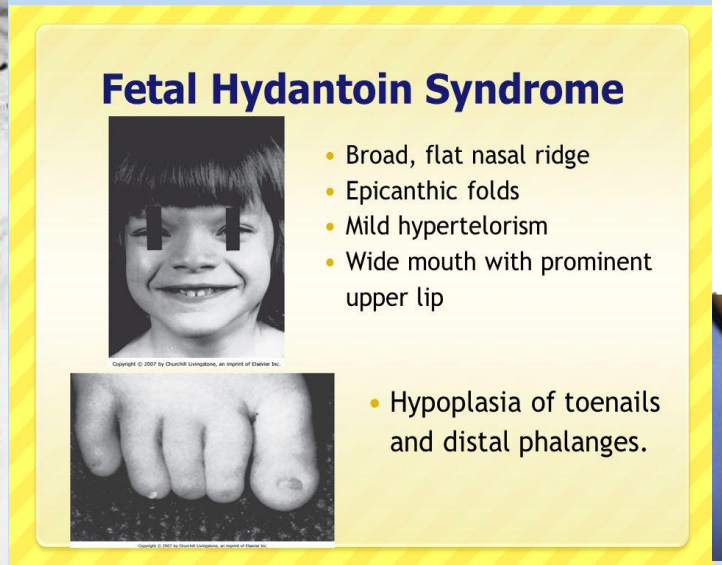
Anencephaly, spina bifida, intellectual disability.



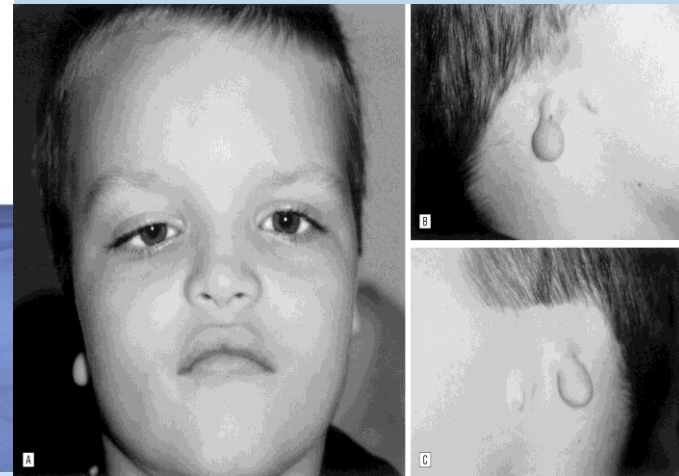
Chemical agents (drugs)

- **Thalidomide**: Limb defects (phocomelia), heart malformations.
- Diphenylhydantoin [phenytoin]:**Fetal hydantoin syndrome**: facial defects, intellectual disability.
- Valproic acid: **Fetal valproate syndrome**: NTDs; heart, craniofacial, and limb anomalies.
- Vitamin A: **Isotretinoin embryopathy**: small, abnormally shaped ears, mandibular hypoplasia, cleft palate, heart defects.









Chemical agents (Alcohol)

- **Fetal alcohol spectrum disorder (FASD).**
- **Fetal alcohol syndrome (FAS)**: IUGR, short palpebral fissures, maxillary hypoplasia, indistinct philtrum, thin upper lip, heart defects, intellectual disability.
- **Alcohol-related neurodevelopmental disorder (ARND).**

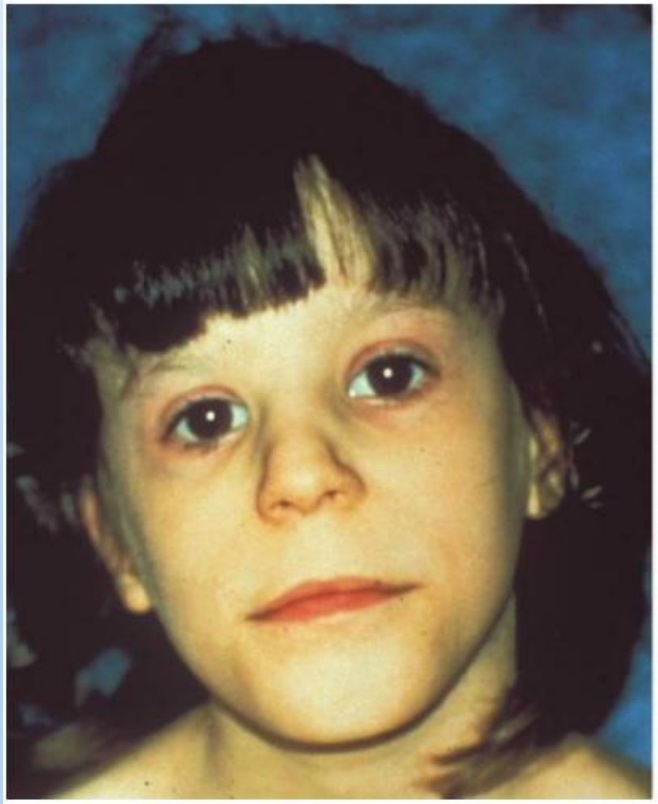
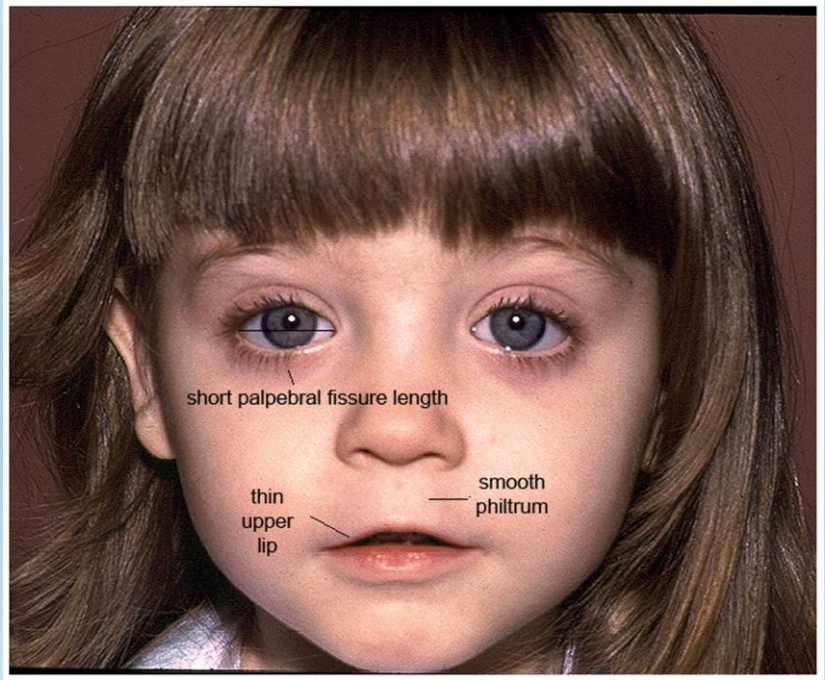
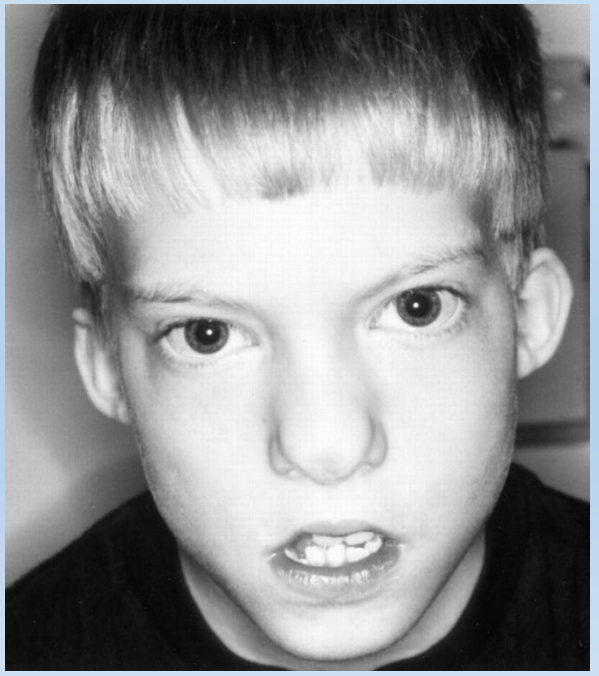


FIGURE 9.6 Characteristic features of a child with FAS, including an indistinct philtrum, thin upper lip, depressed nasal bridge, short nose, and flat midface.





Chemical agents (Heavy metals)

- **Organic mercury**: Neurological symptoms similar to those of cerebral palsy.
- **Lead**: Growth retardation, neurological disorders.

Maternal Disease

- **Diabetes mellitus (DM)**: high incidence of stillbirths, neonatal deaths, abnormally large infants, and congenital malformations.
- The risk of congenital anomalies in children born to mothers with gestational, type 1 and type 2 diabetes is three to four times that for nondiabetic mothers.
- Strict control of maternal glucose levels beginning before conception and continuing throughout gestation reduces the occurrence of malformations.



Figure 27.2 (*Left*) A macrosomic baby born to a mother with diabetes. (*Right*) A normal baby born to a mother without diabetes.

Maternal DM

- Increased risk is for a wide variety of malformations, including neural tube defects and congenital heart defects.
- There is also a higher risk for **caudal dysgenesis (sirenomelia) or the mermaid baby.**
- Altered glucose levels may affect the **BRACHYURY gene** that regulates dorsal mesoderm formation.





Nutritional Deficiencies

- **Endemic cretinism**, caused by **iodine deficiency** and characterized by stunted mental and physical growth.



Thank you

